CSCI 4100
Operating Systems and Architecture

Linux Basics

Topics

- Getting Started
- Utilities
- Files
- Directories

Getting Started

- A **shell** is a command line interface to an operating system.
- If you are running a Linux system on your laptop or desktop, you can access a shell by starting a **terminal** program.
- You can also remotely access a Linux server by using a program that uses the **Secure Shell Protocol (SSH)** to connect to and log in to the server with a username and password.
- Once you have access to a shell it will display a **command prompt** and wait for you to enter a command and type ENTER.
- A Linux command prompt is typically a string of text ending with $.

Finding Files and Directories

- A **directory** is a listing of a set of files, similar to a folder in a graphical user interface.
- To see the current directory, use the `pwd` (print working directory) command.

```
$ pwd
/home/colemann
```

- To list the contents of the current directory, use the `ls` command.

```
$ ls
example scripts source test
```

Commands and Logging Out

- You can use the up and down arrow keys to repeat previous commands and the TAB key to auto complete commands and file names.
- If you are using SSH to access a remote server, you can log out by typing the `exit` command or CONTROL-d.

```
[colemann@apmycs2 ~]$ exit
logout

Connection to apmycs2.apsu.edu closed.
```

Utilities

- Most Linux utilities are programs with short names that can be executed by typing the name of the program after the command prompt.
- Many of these utilities have **options** and/or **arguments** that follow the name.
- For example, the `ls` command on its own lists the contents of the current working directory.

```
$ ls
colors.1 colors.2 days months practice presidents
```

Options

- The `ls` command followed by the `-l` option does the same thing, but in long format, including additional information about each file or directory.

```
$ ls –l
total 24
-rw-r--r-- 1 colemann faculty 29 Sep 1 09:32 colors.1
-rw-r--r-- 1 colemann faculty 22 Sep 1 09:33 colors.2
-rw-r--r-- 1 colemann faculty 57 Sep 1 09:32 days
-rw-r--r-- 1 colemann faculty 48 Sep 1 09:31 months
-rw-r--r-- 1 colemann faculty 56 Sep 1 09:27 practice
-rw-r--r-- 1 colemann faculty 92 Sep 1 09:49 presidents
```

Arguments

- The `ls` command followed by an argument consisting of the name of a file in the directory will list only that file.

```
$ ls practice
practice
```

- If both options and arguments are used, the options should come first.

```
$ ls -l practice
-rw-r--r-- 1 colemann faculty 56 Sep 1 09:27 practice
```

Copying and Removing Files

- The `cp` command copies a file.

```
$ cp practice practice2
$ ls
colors.1 colors.2 days months practice practice2 presidents
```

- The `rm` command removes a file.

```
$ rm practice2
$ ls
colors.1 colors.2 days months practice presidents
```

Displaying Files

- The `cat` command displays the contents of a text file to the terminal.

```
$ cat practice2
This is a small file that I created
with a text editor.
```

- The `more` and `less` commands are similar to `cat`, except they display the contents of a text file one page at a time.
- To go to the next page with `more` or `less`, press the SPACE bar.
- The `less` command also supports using the up and down arrow keys to move through a file line by line.

Making and Removing Directories

- The `mkdir` command makes a new directory.

```
$ mkdir dir1 dir2            # makes two directories
$ ls
dir1 dir2
```

- The `rmdir` command removes an empty directory.

```
$ rmdir dir1                 # removes the dir1 directory
$ ls
dir2
```

Changing Directories

- The `cd` command changes the working directory.

```
$ pwd
/home/ncoleman/example

$ cd dir2                       # changes to the dir2 directory
$ pwd
/home/ncoleman/example/dir2

$ cd ..                         # changes back to the example
$ pwd                           # directory
/home/ncoleman/example
```

Moving and Renaming Files

- The `mv` command can be used to move or rename a file or directory.

```
$ ls
file1.txt file2.txt newdir

$ mv file2.txt newdir          # moves file2.txt inside of newdir
$ ls
file1.txt newdir
$ ls newdir
file2.txt

$ mv file1.txt file3.txt       # renames file1.txt to file3.txt
$ ls
file3.txt newdir
```